

Diagram illustrating the Tucker decomposition of a 3D tensor T into a core tensor G and three factor matrices A , B , and C .

The tensor T is shown as a purple cube, labeled T and $\in \mathbb{R}^{n \times m \times p}$.

The decomposition is represented by the equation:

$$T = \sum_{r=1}^R a_r \otimes b_r \otimes c_r$$

where the components are visualized as follows:

- $a_1, a_2, \dots, a_r$ are blue vertical bars representing the first factor matrices.
- $b_1, b_2, \dots, b_r$ are orange horizontal bars representing the second factor matrices.
- $c_1, c_2, \dots, c_r$ are green diagonal bars representing the third factor matrices.

The diagram shows the tensor T is equal to the sum of the outer products of the factor matrices a_r , b_r , and c_r for $r=1, 2, \dots, r$.